

Supplementary Information: Audited reconstruction of contourlet steganography

Author Name

S1. Provenance chain

Every reported result is linked to the numerical execution commit `7ff0c5abf4511c803a935645dcc2c3ed012f05e9`, the final reporting commit `c564d209844a9dfdb74bf9031fa1ddf3af72cad4`, run ID `f7acf6d9d31dd66cddf1`, and runtime gate `20260730T134046Z-98265862`. The validated result archive contains 1,582 files and 88 immutable objects; its SHA-256 is `5a367ddb07c3df88c2a3ea7ec38187d1ea195e898a61baaa8e733d6dd347b663`.

S2. Reproduction capsule

The private capsule contains the exact input manifests, rights metadata, complete Contourlet tree, environment inventories, offline Git bundle, restore instructions and the validated result archive. Its SHA-256 is `d88a022b2cfca54ab2c9572bd6d0a442a61f169d97c4e27ffe59ef0adccd431d`. Independent matching found zero mismatches for 27 input files, 72 toolbox files and 242 evidence files. Image-derived artifacts are not redistributed publicly.

S3. Locked evaluation matrix

Condition	Methods	Rows
Clean	4 methods $\times$ 4 cases	16
JPEG quality 70	4×4	16
Gaussian variance 10	4×4	16
Salt-and-pepper density 0.03	4×4	16
Core total		64
Triggered hard checks	C0/C3 pairs	24
Overall total		88

S4. Metric definition

For a payload of length B , the effective unrecovered bit rate is

$$\text{EUR} = \frac{1}{B} \sum_{i=1}^B \mathbf{1}[\hat{b}_i \neq b_i].$$

Thus $\text{EUR} = 0$ denotes exact payload recovery and $\text{EUR} = 1$ denotes complete bit failure under the recorded extraction contract. CRC status, layer recovery, PSNR, SSIM, runtime, memory and all failure states remain in the raw evaluation records.

S5. Reproduction procedure

Checkout the recorded code commit, restore the private input and toolbox trees from the capsule, verify all SHA-256 inventories, install the pinned environment, and run the locked configuration without changing the

payload, transform fingerprint, channel definitions or row budget. Regenerated tables must be derived from the raw CSV/JSON/Parquet rows; manual transcription is not part of the protocol.